

How exactly does AI use a ton of water?

Posted by u/ConsistentCan4633 • 0 points • 13 comments

I get that it's for cooling but isn't it a closed loop? Wouldn't it just take a ton of water to fill everything up initially but it wouldn't be a recurring thing other than maintenance?

Comments

u/peterg4567 • 1 points • 2026-03-13 23:28 UTC

Most water figures for AI usage also include the water used to generate the electricity involved. This is often much more than the water used at the actual data center for cooling.

u/AgentElman • 1 points • 2026-03-13 21:38 UTC

There is closed loop cooling and evaporative cooling.

Closed loop cooling uses little water, evaporative cooling uses a lot of water.

It varies by how they built the AI facility.

u/oldmanonanEbike • 5 points • 2026-03-13 21:21 UTC

For cooling. But the supply needs to be constantly refreshed otherwise the water too will heat up.

u/ConsistentCan4633 • 3 points • 2026-03-13 21:23 UTC

Wouldn't the water just go to radiators or something to cool off and then be cycled through again? You wouldn't need to bring in new water?

u/[deleted] • 1 points • 2026-03-13 21:34 UTC

[removed]

u/unic0de000 • 2 points • 2026-03-13 21:31 UTC

You *can* use water as coolant in a closed loop like that, but it's cheaper to just let the hot water fly away as steam, and just suck up more cool water from the river. So that's what they usually do.

u/Green_Campaign_7818 • 1 points • 2026-03-13 21:35 UTC

So how is evaporated water considered a loss when it will turn into clouds and rain back down? That sounds like a closed loop to me

u/unic0de000 • 1 points • 2026-03-13 21:42 UTC

When we talk about wasting water, we're basically never talking about that water *leaving the earth*. It's not lost from the planet, but it's lost from that specific local ecosystem.

The water level of the river or lake gets lower, and this has effects for plant and animal life, and if other people are drawing water from that same source or further downstream, there might not be enough to go around.

u/Jakobites • 1 points • 2026-03-13 21:37 UTC

Because the rain falls somewhere inaccessible to the people that live where it came from originally.

u/Green_Campaign_7818 • 1 points • 2026-03-13 21:40 UTC

So why not just build the ai farms at an ocean or in naturally cold regions

u/Jakobites • 2 points • 2026-03-13 21:44 UTC

Far from existing power plants and transmission infrastructure for the large amounts of power needed.

Fix's one problem but creates a new more costly problem.

u/ConsistentCan4633
• 1 points •
2026-03-13
22:12 UTC

Maybe if
there's
progress on
more
economical
modular
nuclear
reactors
having data
centers in
cold regions
could work?